

Engineering Mechanics: ME101

Lecture 12-13: Statics

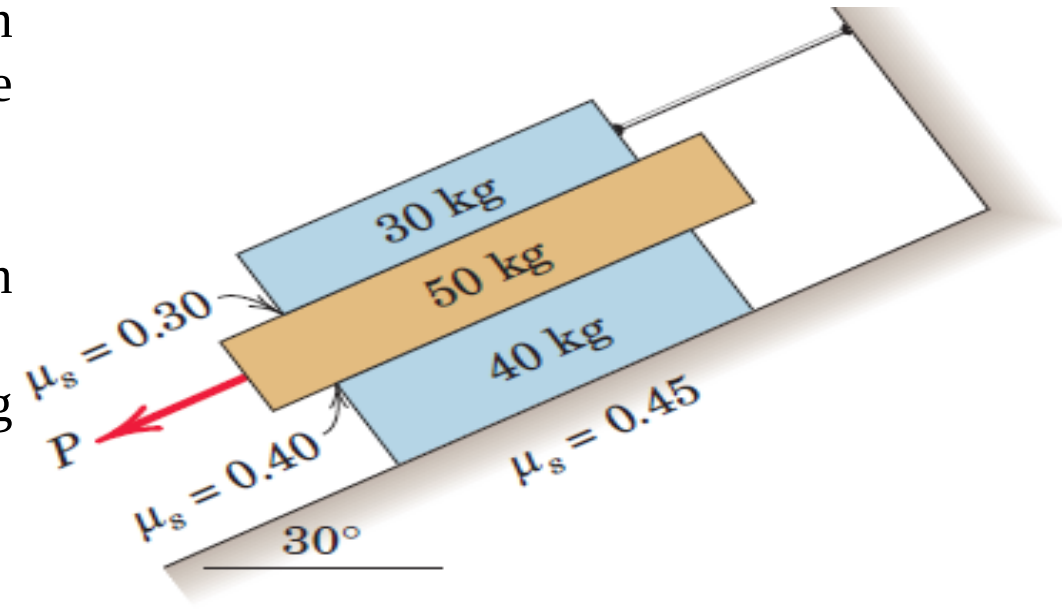


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(Problem-4) The three flat blocks are positioned on the 30 deg. incline as shown in **Fig(c)** , and a force P parallel to the incline is applied to the middle block. The upper block is prevented from moving by a wire which attaches it to the fixed support. The coefficient of static friction for each of the three pairs of mating surfaces is shown. Determine the maximum value which P may have before any slipping takes place?

Solution:

- ✓ The free-body diagram of each block is drawn. The friction forces are assigned in the directions to oppose the relative motion which would occur if no friction were present.
- ✓ There are two possible conditions for impending motion.
- ✓ Either the 50-kg block slips and the 40-kg block remains in place, or
- ✓ The 50- and 40-kg blocks move together with slipping occurring between the 40-kg block and the incline.



Fig(c)

Problem 4 (contd.)

- ✓ The normal forces, which are in the y -direction, may be determined without reference to the friction forces, which are all in the x -direction.

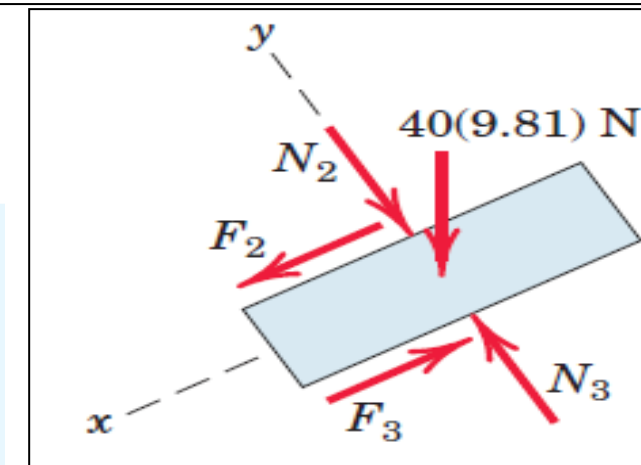
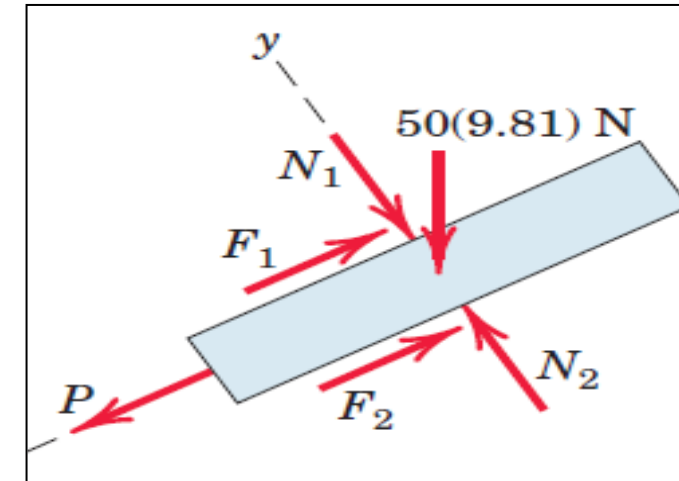
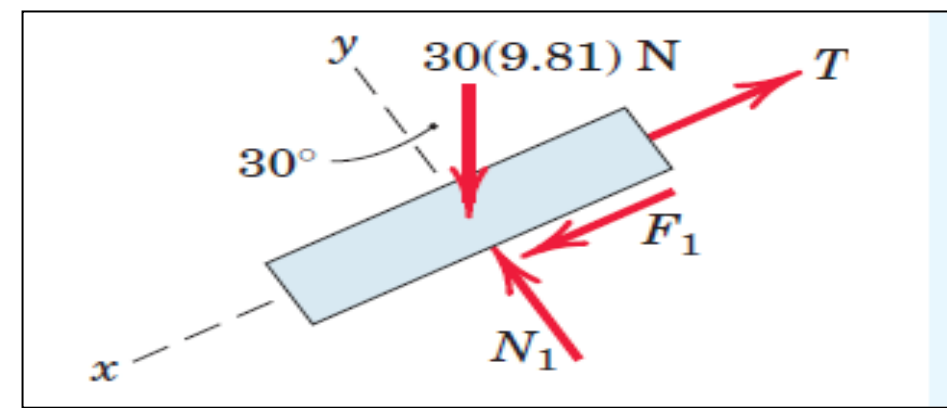
$$\begin{array}{llll}
 [\Sigma F_y = 0] & (30\text{-kg}) & N_1 - 30(9.81) \cos 30^\circ = 0 & N_1 = 255 \text{ N} \\
 & (50\text{-kg}) & N_2 - 50(9.81) \cos 30^\circ - 255 = 0 & N_2 = 680 \text{ N} \\
 & (40\text{-kg}) & N_3 - 40(9.81) \cos 30^\circ - 680 = 0 & N_3 = 1019 \text{ N}
 \end{array}$$

- ✓ We will assume arbitrarily that only the 50-kg block slips, so that the 40-kg block remains in place. Thus, for impending slippage at both surfaces of the 50-kg block, we have

$$[F_{\max} = \mu_s N] \quad F_1 = 0.30(255) = 76.5 \text{ N} \quad F_2 = 0.40(680) = 272 \text{ N}$$

The assumed equilibrium of forces at impending motion for the 50-kg block gives

$$[\Sigma F_x = 0] \quad P - 76.5 - 272 + 50(9.81) \sin 30^\circ = 0 \quad P = 103.1 \text{ N}$$



Problem 4 (contd.)

- ✓ We now check on the validity of our initial assumption. For the 40-kg block with $F_2 = 272$ N the friction force F_3 would be given by,

$$[\Sigma F_x = 0] \quad 272 + 40(9.81) \sin 30^\circ - F_3 = 0 \quad F_3 = 468 \text{ N}$$

But the maximum possible value of

$$F_3 \text{ is } F_3 = \mu_s N_3 = 0.45(1019) = 459 \text{ N.}$$

- ✓ Thus, 468 N cannot be supported and our initial assumption was wrong. We conclude, therefore, that slipping occurs first between the 40-kg block and the incline. With the corrected value $F_3 = 459$ N, equilibrium of the 40-kg block for its impending motion, requires,

$$[\Sigma F_x = 0] \quad F_2 + 40(9.81) \sin 30^\circ - 459 = 0 \quad F_2 = 263 \text{ N}$$

Equilibrium of the 50-kg block gives, finally,

$$[\Sigma F_x = 0] \quad P + 50(9.81) \sin 30^\circ - 263 - 76.5 = 0$$

$$P = 93.8 \text{ N}$$

Ans.

Thus, with $P = 93.8$ N, motion impends for the 50-kg and 40-kg blocks as a unit.

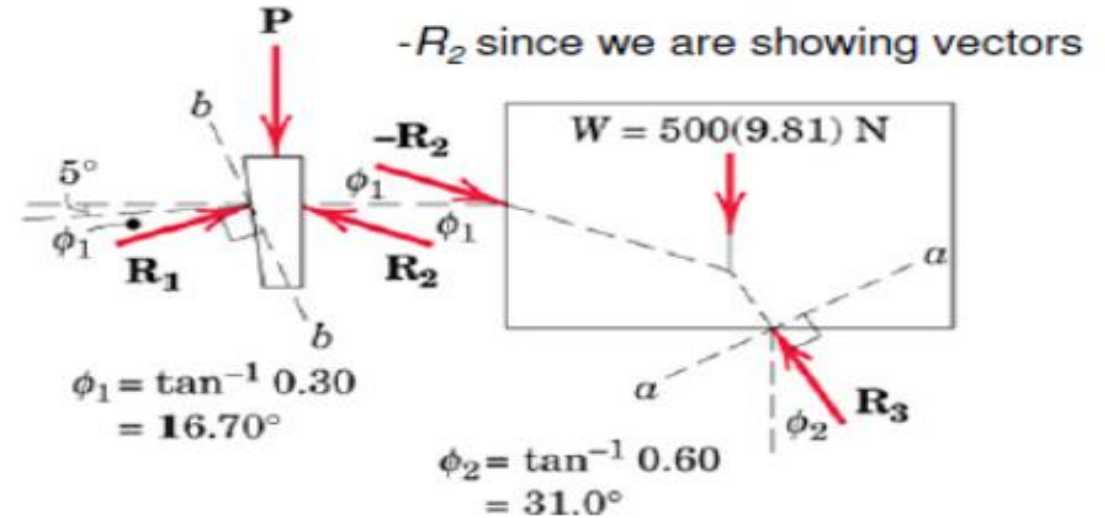
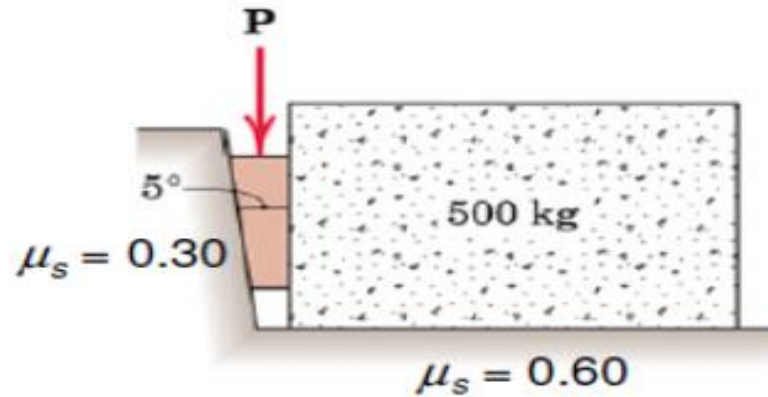
Problem 5

Coefficient of Static Friction for both pairs of wedge = 0.3

Coefficient of Static Friction between block and horizontal surface = 0.6

Find the least P required to move the block

Solution: Draw FBDs



Problem 5 (contd.)

Solution: $w = 500 \times 9.81 = 4905 \text{ N}$

Three ways to solve

Method 1:

Equilibrium of FBD of the Block

$$\sum F_x = 0$$

$$R_2 \cos \phi_1 = R_3 \sin \phi_2 \rightarrow R_2 = 0.538 R_3$$

$$\sum F_y = 0$$

$$4905 + R_2 \sin \phi_1 = R_3 \cos \phi_2 \rightarrow R_3 = 6970 \text{ N}$$

$$\rightarrow R_2 = 3750 \text{ N}$$

Equilibrium of FBD of the Wedge

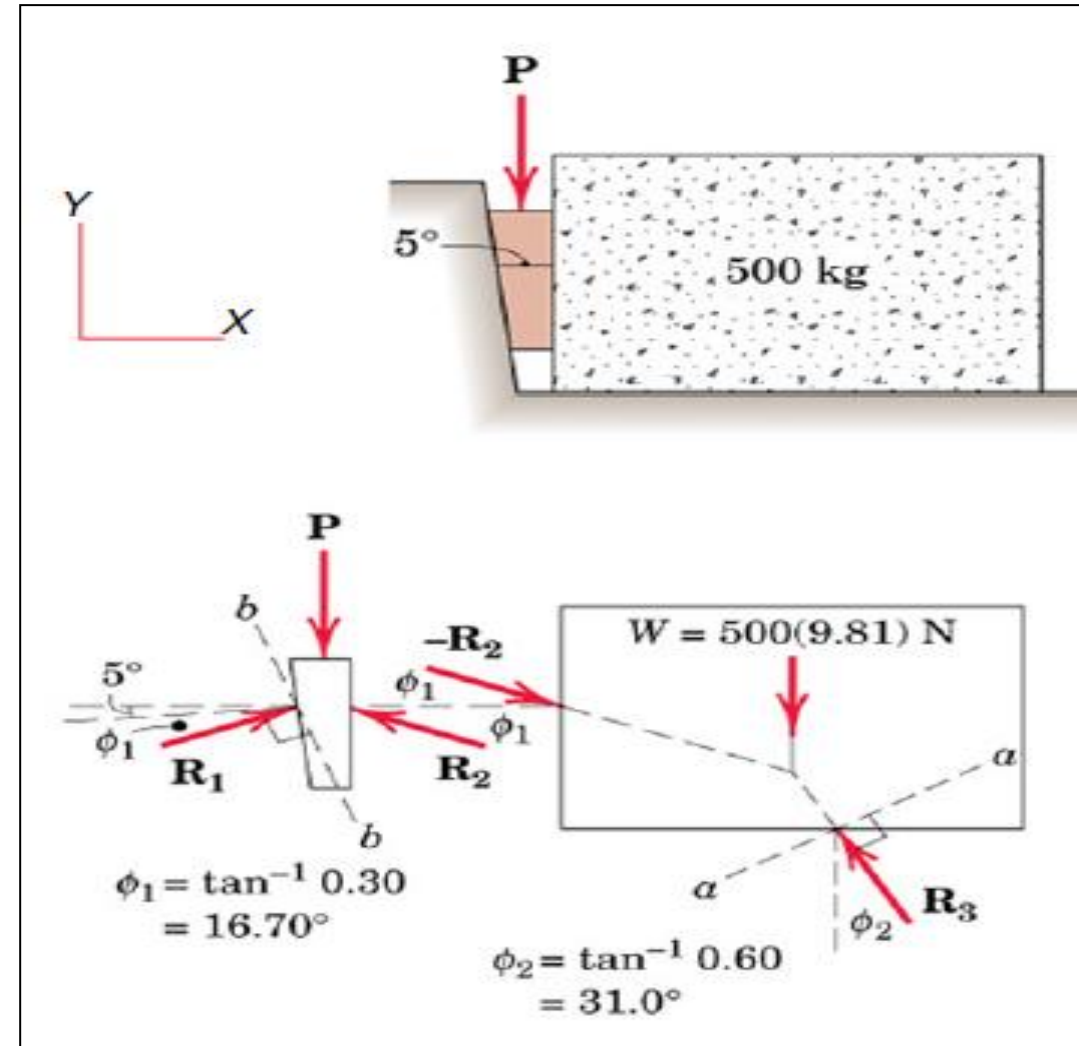
$$\sum F_x = 0$$

$$R_2 \cos \phi_1 = R_1 \cos(\phi_1 + 5) \rightarrow R_1 = 3871 \text{ N}$$

$$\sum F_y = 0$$

$$R_1 \sin(\phi_1 + 5) + R_2 \sin \phi_1 = P$$

$$\rightarrow P = 2500 \text{ N}$$



Problem 5 (contd.)

Method 2:

Using Equilibrium equations along reference axes *a-a* and *b-b*
→ No need to solve simultaneous equations

Angle between R_2 and *a-a* axis = $16.70 + 31.0 = 47.7^\circ$

Equilibrium of Block:

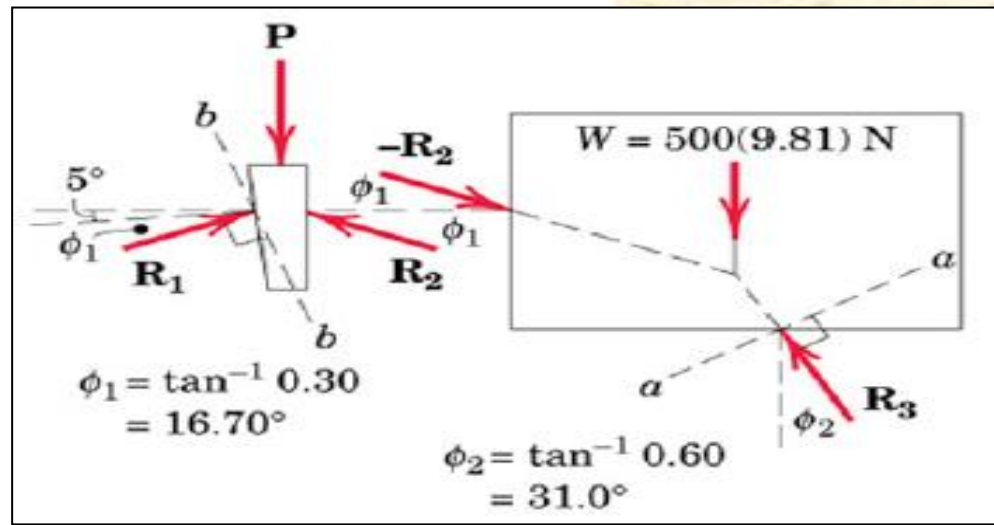
$$[\Sigma F_a = 0] \quad 500(9.81) \sin 31.0^\circ - R_2 \cos 47.7^\circ = 0$$
$$R_2 = 3750 \text{ N}$$

Equilibrium of Wedge:

Angle between R_2 and *b-b* axis = $90 - (2\phi_1 + 5) = 51.6^\circ$

Angle between P and *b-b* axis = $\phi_1 + 5 = 21.7^\circ$

$$[\Sigma F_b = 0] \quad 3750 \cos 51.6^\circ - P \cos 21.7^\circ = 0$$
$$P = 2500 \text{ N}$$



Problem 5 (contd.)

Method 3:

Graphical solution using vector polygons

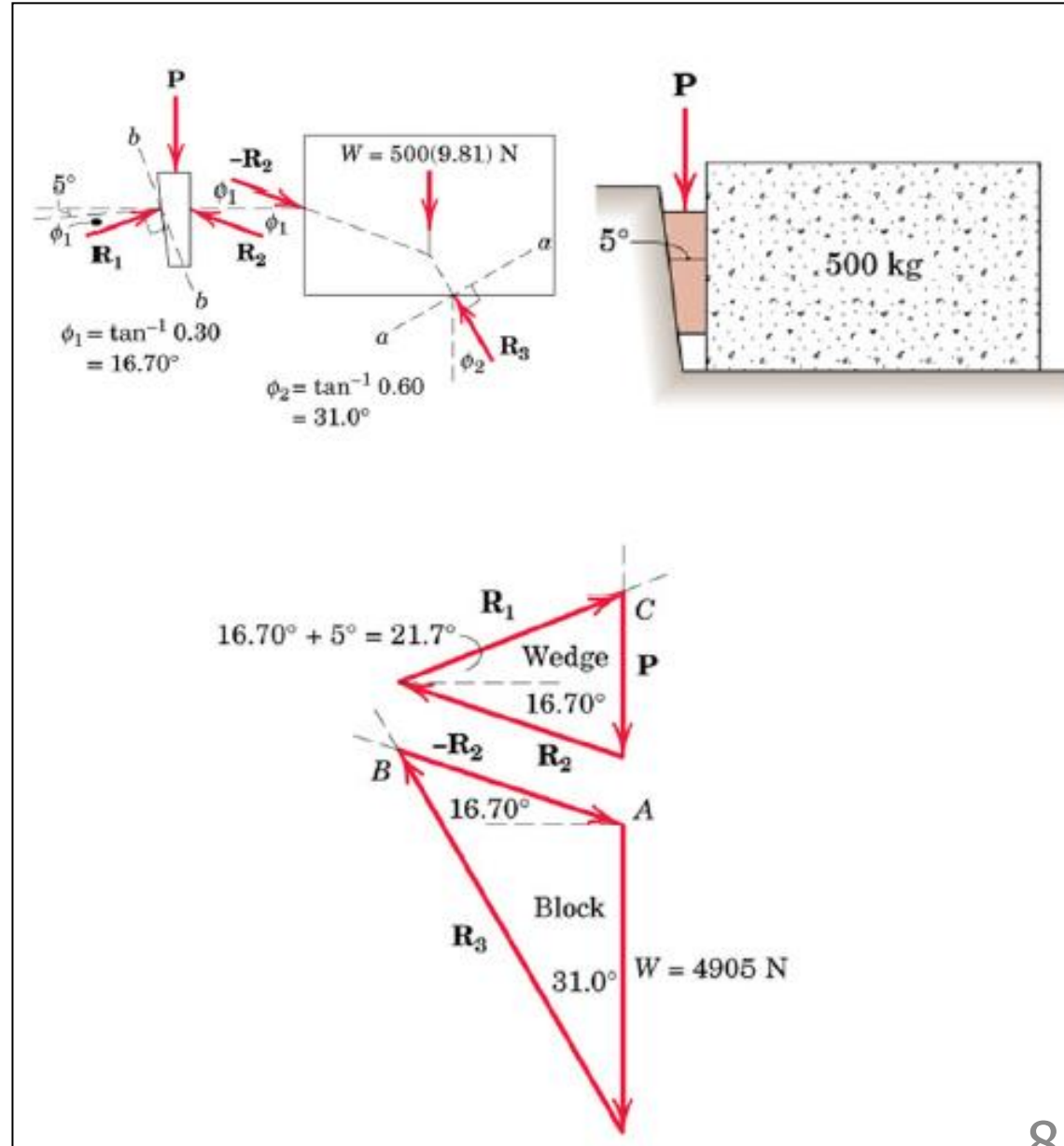
Starting with equilibrium of the block:

W is known, and directions of R_2 and R_3 are known

→ Magnitudes of R_2 and R_3 can be determined graphically

Similarly, construct vector polygon for the wedge from known magnitude of R_2 , and known directions of R_2 , R_1 , and P .

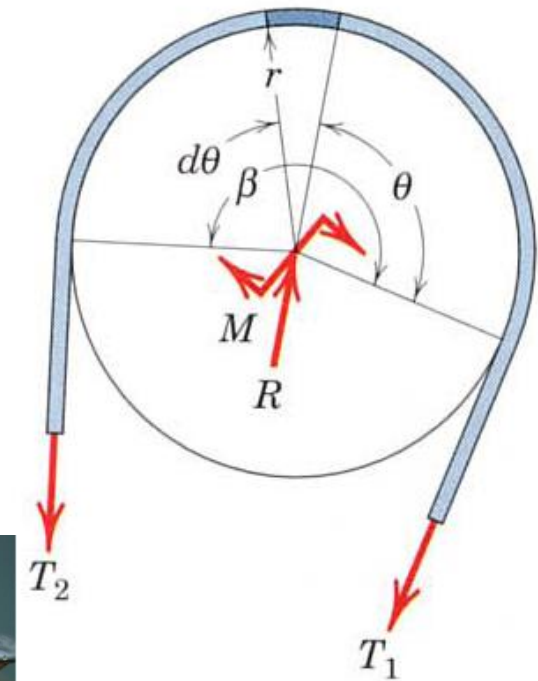
→ Find out the magnitude of P graphically



Belt Friction

(relationship between belt tensions)

- ❖ Impending slippage of flexible cables, belts, ropes over sheaves, wheels, drums
- ✓ It is necessary to estimate the frictional forces developed between the belt and its contacting surface.
 - ❖ Consider a drum subjected to two belt tensions (T_1 and T_2).
 - ✓ M is the torque necessary to prevent rotation of the drum.
 - ✓ R is the bearing reaction.
 - ✓ r is the radius of the drum.
 - ✓ β is the total contact angle between belt and surface. (in radians).
 - ✓ $T_2 > T_1$ since M is clockwise.

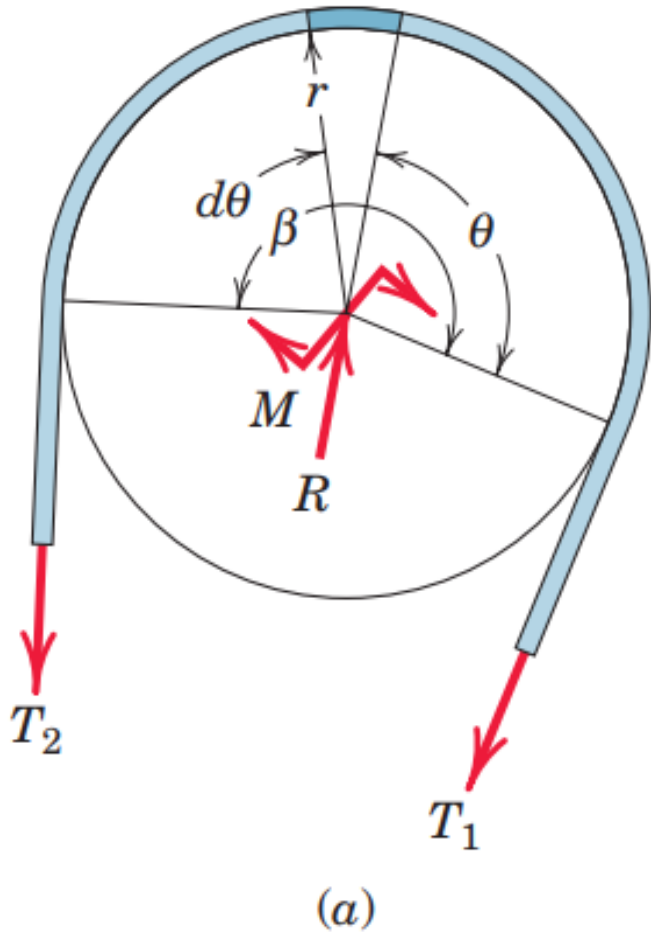


Introduction



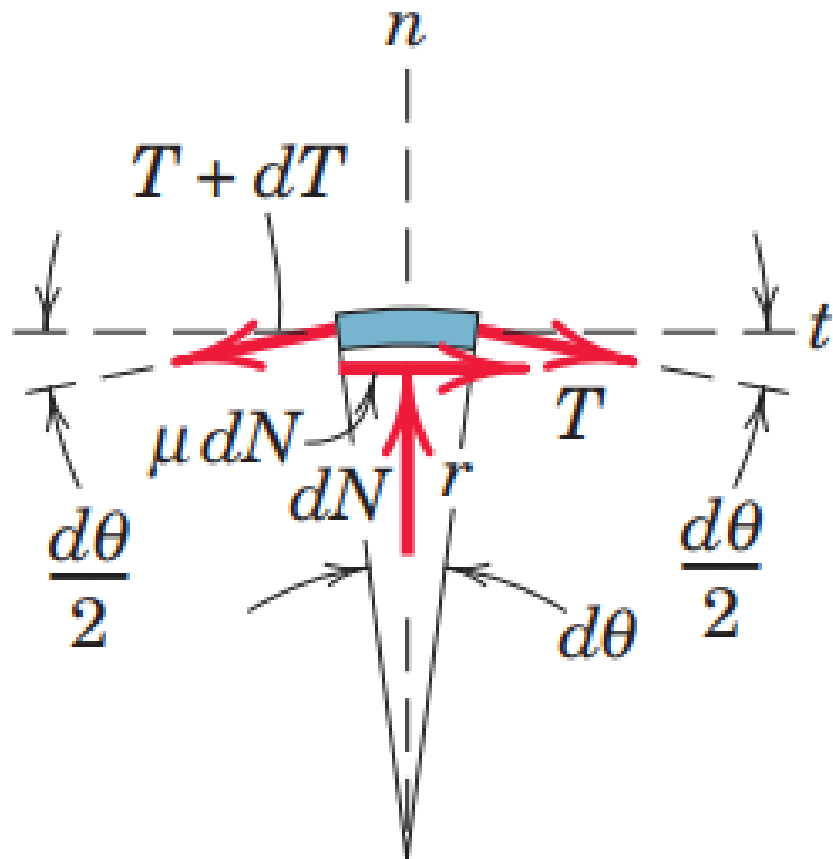
- Concept of Belt friction is extensively used in Belt Drives.
- Just one turn of rope around pulley, **can increase tension by large magnitude.**
- Hence, Tensions in the ropes are analysed by the concept of Belt friction.
- Designer has to make sure that Belt tension does not exceed the Permissible limits to avoid system failure.

Analysis of Belt Friction



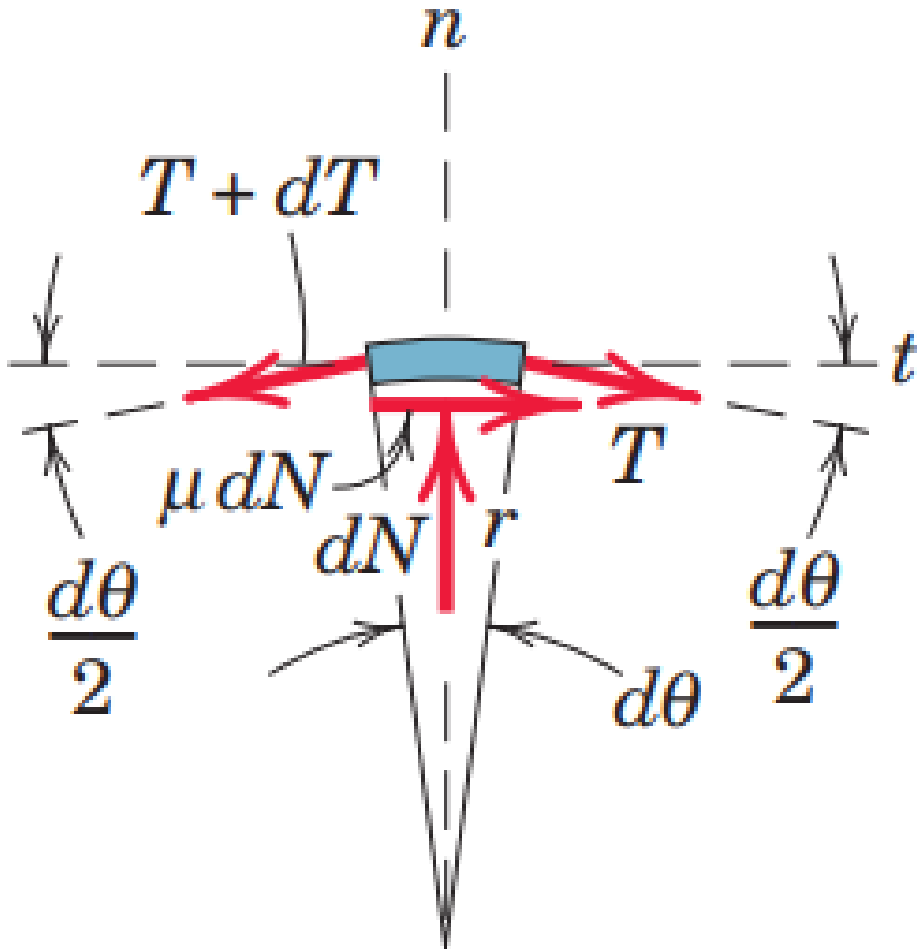
- A Drum is wrapped with a Belt as shown
- It is subjected to two Belt Tensions T_1 and T_2
- Torque M necessary to prevent rotation
- Logically, with M in the CW direction shown, T_2 is greater than T_1 .
- The tension increases from T at the angle θ to $T + dT$ at the angle $\theta + d\theta$ in element

Analysis of Belt Friction



- The tension increases from T at the angle θ to $T + dT$ at the angle $\theta + d\theta$ in element
- The **free body diagram** of an element of the belt of length “ $rd\theta$ ” is shown
- The normal force is a differential dN , since it acts on a differential element of area
- Friction force, which must act on the belt in a direction to oppose slipping, is a differential and is μdN

Analysis of Belt Friction



❖ Force balance in **t-direction** is

$$T \cos \frac{d\theta}{2} + \mu dN = (T + dT) \cos \frac{d\theta}{2}$$

$$\mu dN = dT \quad \dots \dots \dots (1)$$

❖ Force Balance in **n-direction**

$$dN = (T + dT) \sin \frac{d\theta}{2} + T \sin \frac{d\theta}{2}$$

$$dN = T d\theta \quad \dots \dots \dots (2)$$

• Combining equation (1) and (2),

$$\frac{dT}{T} = \mu d\theta$$

Analysis of Belt Friction

$$\frac{dT}{T} = \mu d\theta$$

- Integrating between corresponding limits yields

$$\int_{T_1}^{T_2} \frac{dT}{T} = \int_0^{\beta} \mu d\theta = \ln \frac{T_2}{T_1} = \mu\beta$$

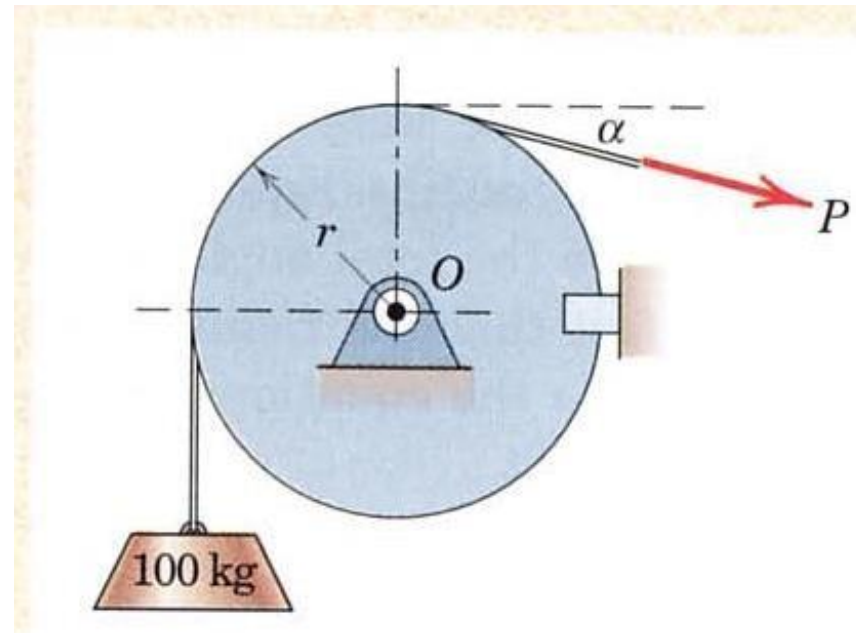


$$T_2 = T_1 e^{\mu\beta}$$

- ✓ Here, " β " is called as **Angle of Wrap** or Total angle of Belt contact.
- ✓ For “n” rotations of belt over pulley, angle of contact will be $2n\pi + \beta$

Sample Problem

A flexible cable which supports the 100-kg load is passed over a fixed circular drum and subjected to a force P to maintain equilibrium. The coefficient of static friction μ between the cable and the fixed drum is 0.30. (a) For $\alpha = 0$, determine the maximum and minimum values which P may have in order not to raise or lower the load. (b) For $P = 500$ N, determine the minimum value which the angle α may have before the load begins to slip.



Solution on next page

Solution of Sample Problem

❖ Impending slippage of the cable over the fixed drum is given by: $T_2 = T_1 e^{\mu\beta}$

1 (a) With $\alpha = 0$ the angle of contact is $\beta = \pi/2$ rad. For impending upward motion of the load, $T_2 = P_{\max}$, $T_1 = 981$ N, and we have

2
$$P_{\max}/981 = e^{0.30(\pi/2)} \quad P_{\max} = 981(1.602) = 1572 \text{ N} \quad \text{Ans.}$$

For impending downward motion of the load, $T_2 = 981$ N and $T_1 = P_{\min}$. Thus,

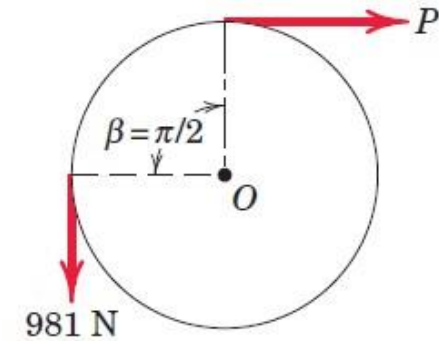
$$981/P_{\min} = e^{0.30(\pi/2)} \quad P_{\min} = 981/1.602 = 612 \text{ N} \quad \text{Ans.}$$

(b) With $T_2 = 981$ N and $T_1 = P = 500$ N, Eq. 6/7 gives us

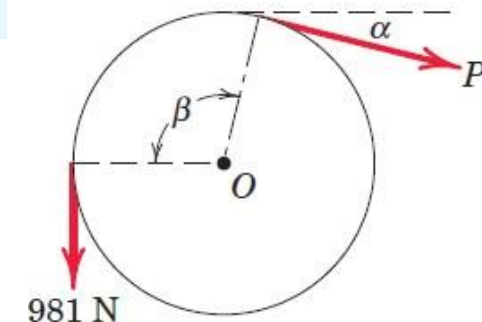
$$981/500 = e^{0.30\beta} \quad 0.30\beta = \ln(981/500) = 0.674$$

$$\beta = 2.25 \text{ rad} \quad \text{or} \quad \beta = 2.25 \left(\frac{360}{2\pi} \right) = 128.7^\circ$$

$$\alpha = 128.7^\circ - 90^\circ = 38.7^\circ$$



(a) $\alpha = 0$



(b) $P = 500$ N

FBD

Wheel Friction or Rolling Resistance

- ❑ Resistance of a wheel to roll over a surface is caused by deformation between two materials of contact.
- ✓ This resistance is not due to tangential frictional forces
- ✓ Entirely different phenomenon from that of dry friction



Steel is very stiff
Low Rolling Resistance



Significant Rolling Resistance
between rubber tyre and tar road



Large Rolling Resistance
due to wet field

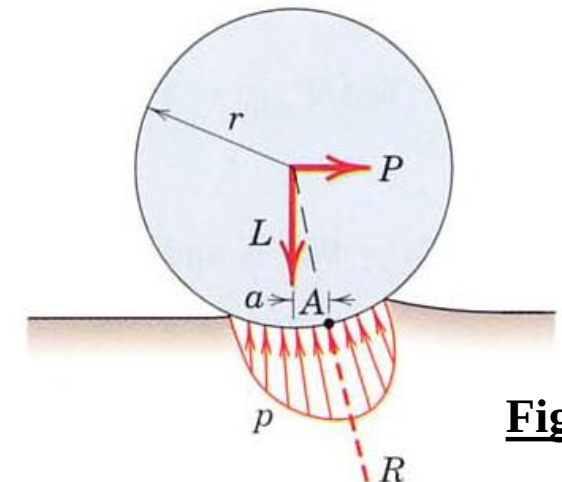
Deformations due to Rolling Resistance

- ❖ Actually materials are not rigid and deformation occurs at the point of contact.
- ✓ Reaction of surface on the cylinder consists of a distribution of normal pressure.
- ❑ Consider a wheel under action of a **load L on axle and a force P applied at its center to produce rolling;**
- ✓ Deformation of wheel and supporting surface as shown in **Fig(f)**
- ✓ Resultant R of the distribution of normal pressure must pass through wheel center for the wheel to be in equilibrium (i.e., rolling at a constant speed).
- ✓ R acts at point A on right of wheel center for rightwards motion.
- ✓ Force P required to maintain rolling at constant speed can be apprx. estimated as:

$$\sum M_A = 0 \Rightarrow L a = P r \cos \theta$$

$$\Rightarrow P = \frac{a}{r} L = \mu_r L$$

- μ_r is the ratio of resisting force to the normal force analogous to μ_s or μ_k and called **Coefficient of Rolling Resistance**
- No slippage or impending slippage in interpretation of μ_r

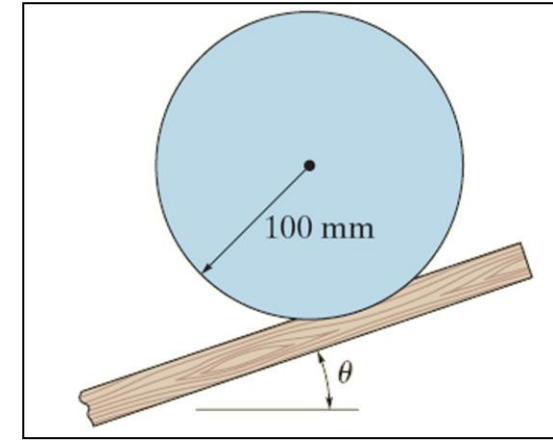


Fig(f)

Note: 3 force member equilibrium condition (mine)

Problem on Rolling Resistance

- A 10 kg steel wheel (radius = 100 mm) rests on an inclined plane made of wood. At $\theta = 1.2^\circ$, the wheel begins to roll-down the incline with constant velocity. Determine the coefficient of rolling resistance in Fig(g)?



Fig(g)

Solution:

- When the wheel has impending motion, the normal reaction N acts at point A defined by the dimension a . Draw the FBD for the wheel: $r = 100$ mm, 10 kg = 98.1 N

□ Using simplified equation directly: $P = \frac{a}{r} L = \mu_r L$

Here $P = 98.1(\sin 1.2) = 2.05$ N

$L = 98.1(\cos 1.2) = 98.08$ N,

❖ Coefficient of Rolling Resistance, $\mu_r = 0.0209$

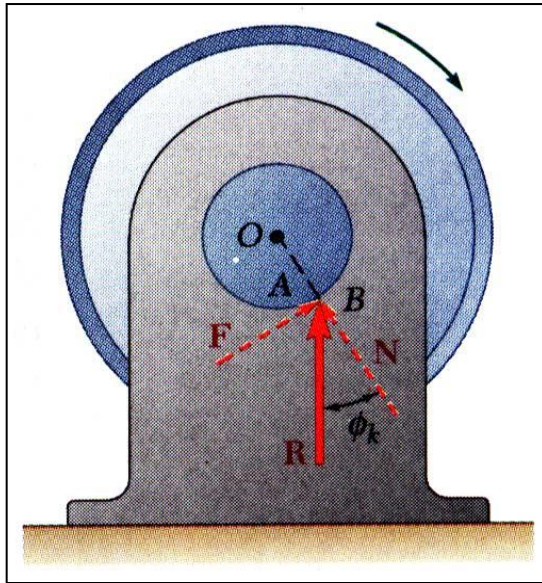
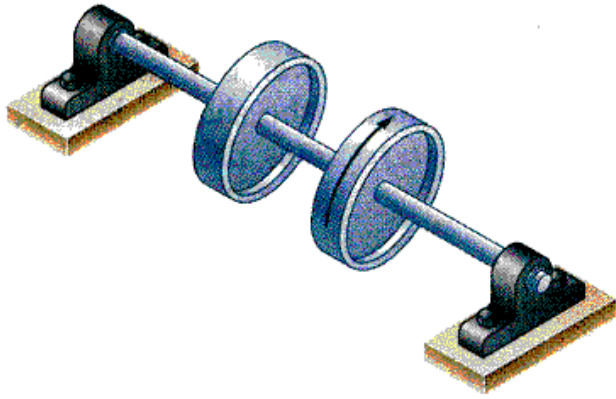
❖ Alternatively, $\sum M_A = 0$

$98.1(\sin 1.2)(r \text{ appx}) = 98.1(\cos 1.2)a$

(since $r \cos 1.2 = r \times 0.9998 \approx r$)

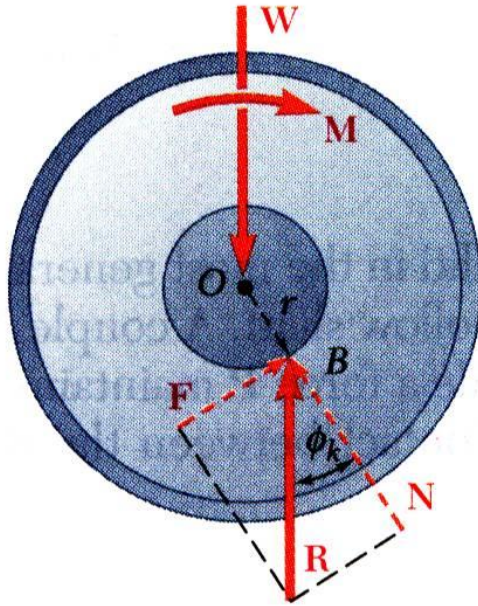
$a/r = \mu_r = 0.0209$

Journal Bearings & Axle Friction



- ✓ **Journal bearings** provide lateral support to rotating shafts. Thrust bearings provide **axial support**.
- ✓ Frictional resistance of fully lubricated bearings depends on clearances, speed and lubricant viscosity. Partially lubricated axles and bearings can be assumed to be in direct contact along a straight line.
- ✓ Forces acting on bearing are weight W of wheels and shaft, couple M to maintain motion, and reaction R of the bearing.
- ✓ Reaction is vertical and equal in magnitude to W .
- ✓ Reaction line of action does not pass through shaft center O ; R is located to the right of O , resulting in a moment that is balanced by M .
- ✓ Physically, contact point is displaced as axle “climbs” in bearing.

Journal Bearings & Axle Friction (*contd.*)

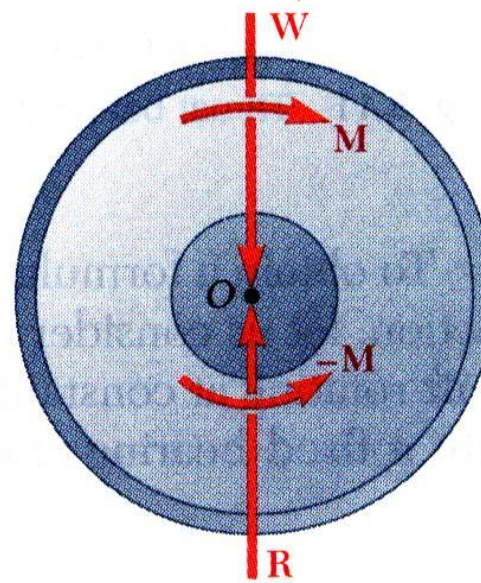


- Angle between R and normal to bearing surface is the angle of kinetic friction ϕ_k .

$$M = Rr \sin \phi_k$$

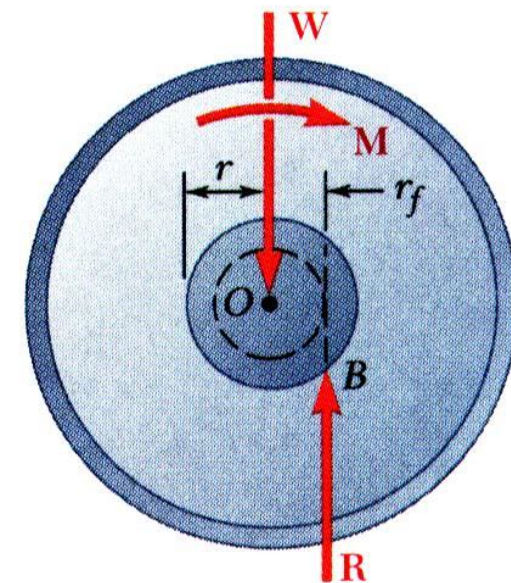
$$\approx Rr \mu_k$$

(1)



- May treat bearing reaction as force-couple system.

(2)



- For graphical solution, R must be tangent to *circle of friction*.

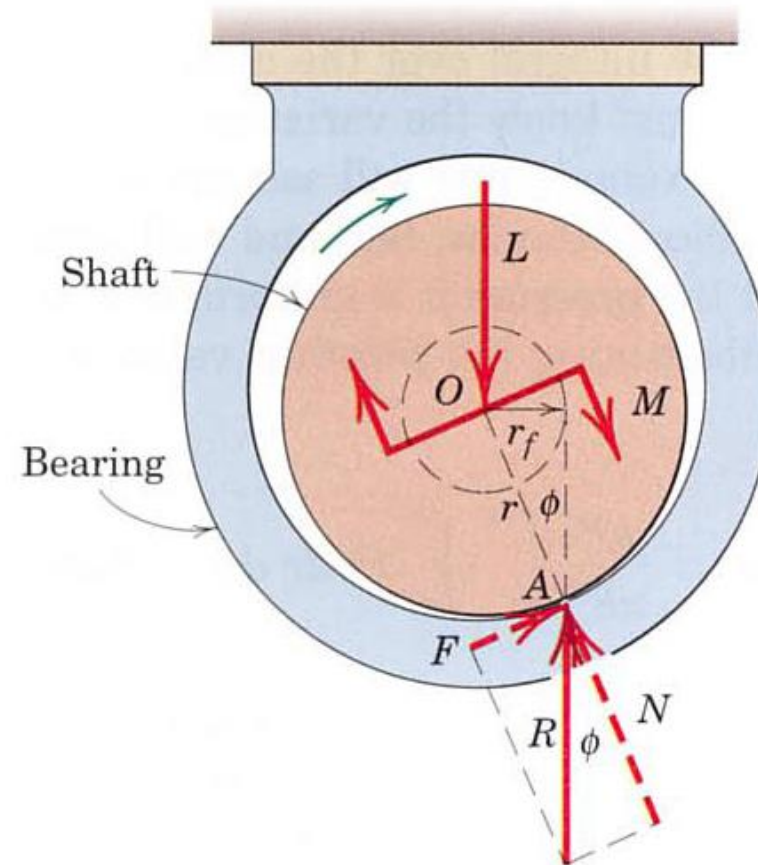
$$r_f = r \sin \phi_k$$

$$\approx r \mu_k$$

Note: (1) Equivalent to (2)

Moment acting on Journal Bearing

$$M = Lr_f = Lr \sin \phi$$



For a small coefficient of friction, the angle ϕ is small, and the sine and tangent may be interchanged with only small error. Since $\mu = \tan \phi$, a good approximation to the torque is

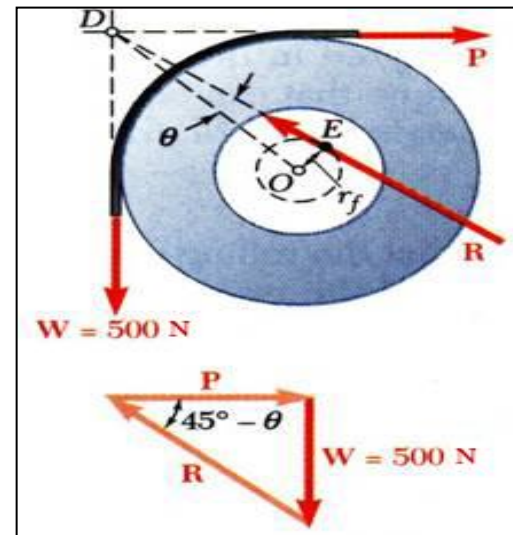
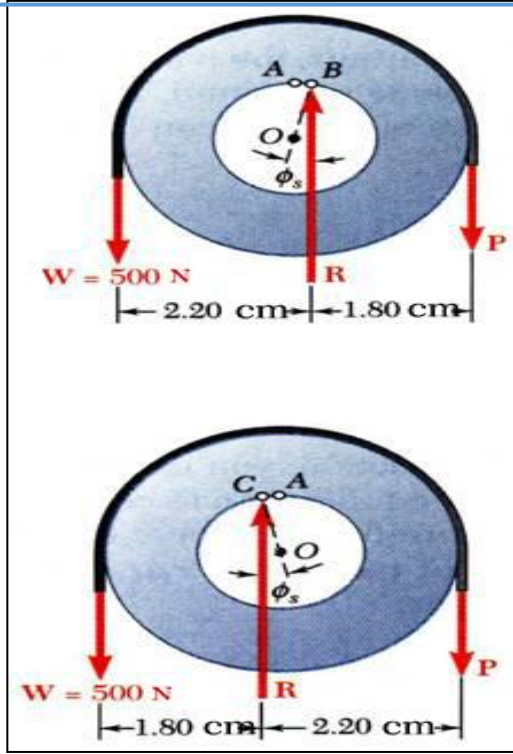
$$M = \mu Lr$$

(6/4a)

Problem on Belt Friction

❖ A pulley of diameter 4 cm can rotate about a fixed shaft of diameter 2 cm. The coefficient of static friction between the pulley and shaft is 0.20. **Determine:**

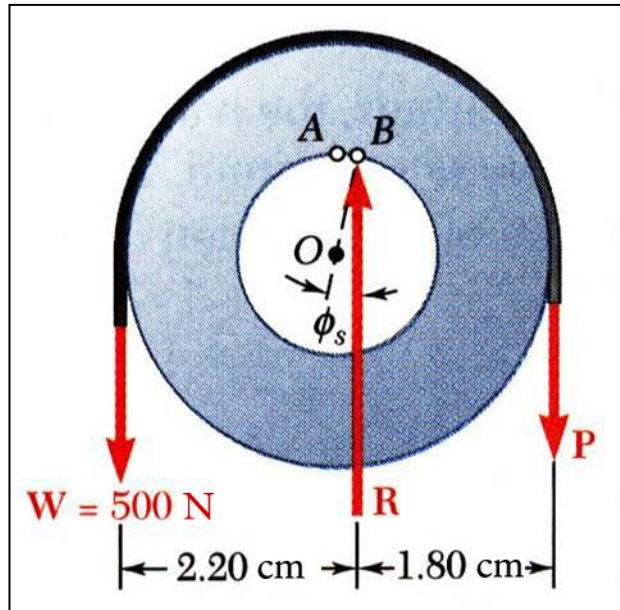
- ✓ the smallest vertical force P required to start raising a 500 N load,
- ✓ the smallest vertical force P required to hold the load, and
- ✓ the smallest horizontal force P required to start raising the same load.



Solution:

- ✓ With the load on the left and force P on the right, impending motion is clockwise to raise load. Sum moments about displaced contact point B to find P .
- ✓ Impending motion is counter-clockwise as load is held stationary with smallest force P . Sum moments about C to find P .
- ✓ With the load on the left and force P acting horizontally to the right, impending motion is clockwise to raise load. Utilize a force triangle to find P .

Problem on Belt Friction (*contd.*)



SOLUTION:

- With the load on the left and force P on the right, impending motion is clockwise to raise load. Sum moments about displaced contact point B to find P .
- The perpendicular distance from center O of pulley to line of action of R is

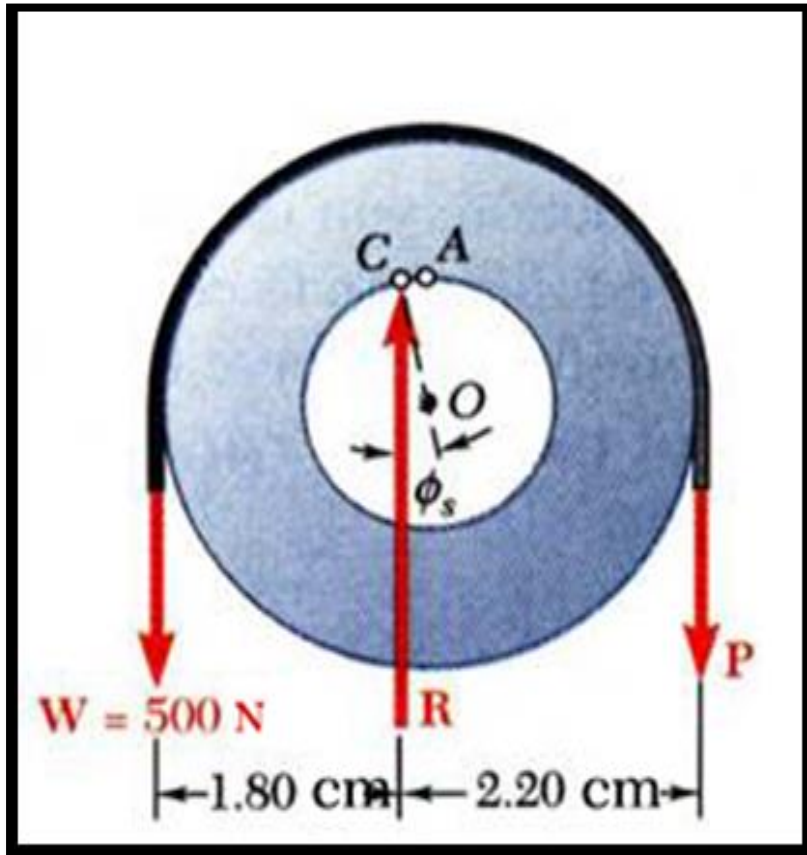
$$r_f = r \sin \phi_s \approx r \mu_s \quad r_f \approx (1 \text{ cm}) 0.20 = 0.20 \text{ cm}$$

- Summing moments about B ,

$$\Sigma M_B = 0: \quad (2.20 \text{ cm})(500 \text{ N}) - (1.80 \text{ cm})P = 0$$

$$P = 611 \text{ N}$$

Problem on Belt Friction (*contd.*)



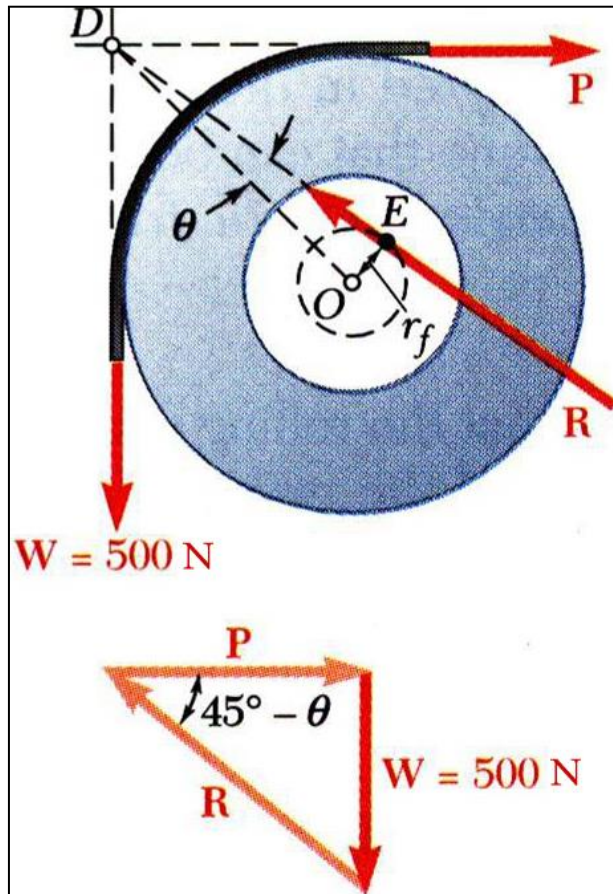
➤ Impending motion is counter-clockwise as load is held stationary with smallest force P . Sum moments about C to find P .

➤ The perpendicular distance from center O of pulley to line of action of R is again 0.20 cm . Summing moments about C ,

$$\Sigma M_C = 0: \quad (1.80\text{ cm})(500\text{ N}) - (2.20\text{ cm})P = 0$$

$$P = 409\text{ N}$$

Problem on Belt Friction (*contd.*)



➤ With the load on the left and force P acting horizontally to the right, impending motion is clockwise to raise load. Utilize a force triangle to find P .

➤ Since W , P , and R are not parallel, they must be concurrent. Line of action of R must pass through intersection of W and P and be tangent to circle of friction which has radius $r_f = 0.20\text{ cm}$.

$$\sin \theta = \frac{OE}{OD} = \frac{0.20\text{ cm}}{(2\text{ cm})\sqrt{2}} = 0.0707$$

$$\theta = 4.1^\circ \quad P = W \cot(45^\circ - \theta) = (500\text{ N}) \cot 40.9^\circ$$

➤ From the force triangle,

$$P = 577\text{ N}$$

Chapter review (contd.)

Screws

Square-threaded screws are used to move heavy loads. They represent an inclined plane, wrapped around a cylinder.

The moment needed to turn a screw depends upon the coefficient of friction and the screw's lead angle θ .

If the coefficient of friction between the surfaces is large enough, then the screw will support the load without tending to turn, i.e., it will be self-locking.

$$M = rW \tan(\theta + \phi_s)$$

Upward Impending Screw Motion

$$M' = rW \tan(\theta - \phi_s)$$

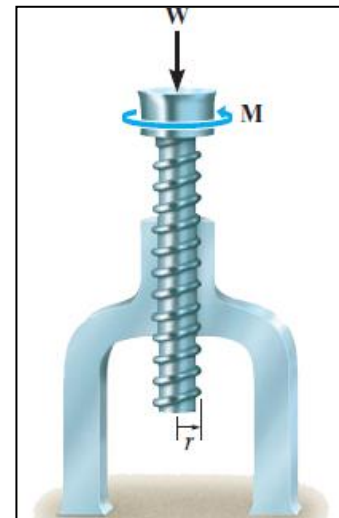
Downward Impending Screw Motion

$$\theta > \phi_s$$

$$M'' = rW \tan(\phi_s - \theta)$$

Downward Screw Motion

$$\phi_s > \theta$$



End

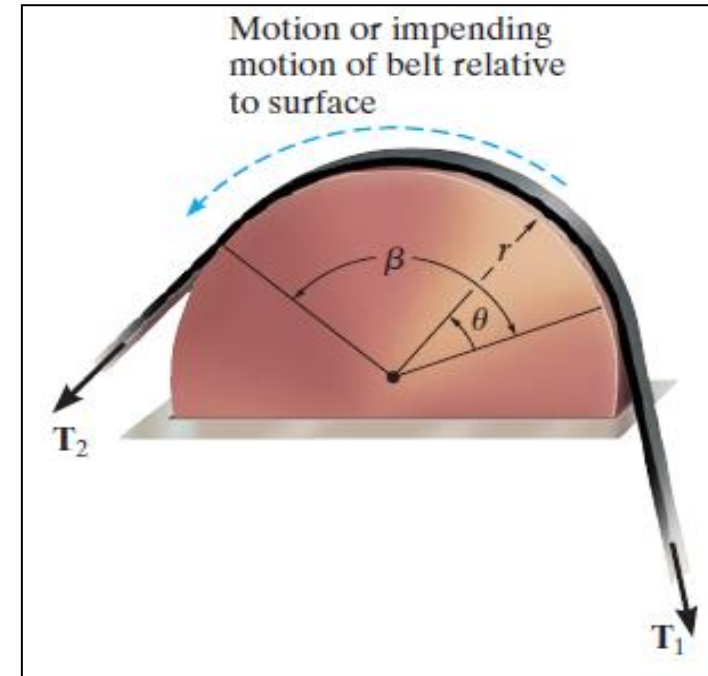
Chapter review (contd.)

Flat Belts

The force needed to move a flat belt over a rough curved surface depends only on the angle of belt contact, β , and the coefficient of friction.

$$T_2 = T_1 e^{\mu\beta}$$

$$T_2 > T_1$$

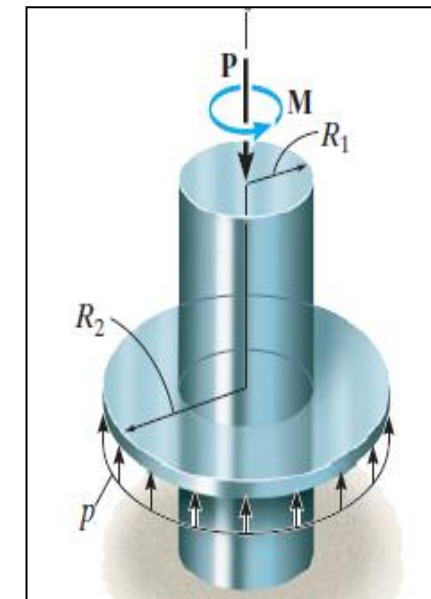


Collar Bearings and Disks

The frictional analysis of a collar bearing or disk requires looking at a differential element of the contact area. The normal force acting on this element is determined from force equilibrium along the shaft, and the moment needed to turn the shaft at a constant rate is determined from moment equilibrium about the shaft's axis.

If the pressure on the surface of a collar bearing is uniform, then integration gives the result shown.

$$M = \frac{2}{3} \mu_s P \left(\frac{R_2^3 - R_1^3}{R_2^2 - R_1^2} \right)$$



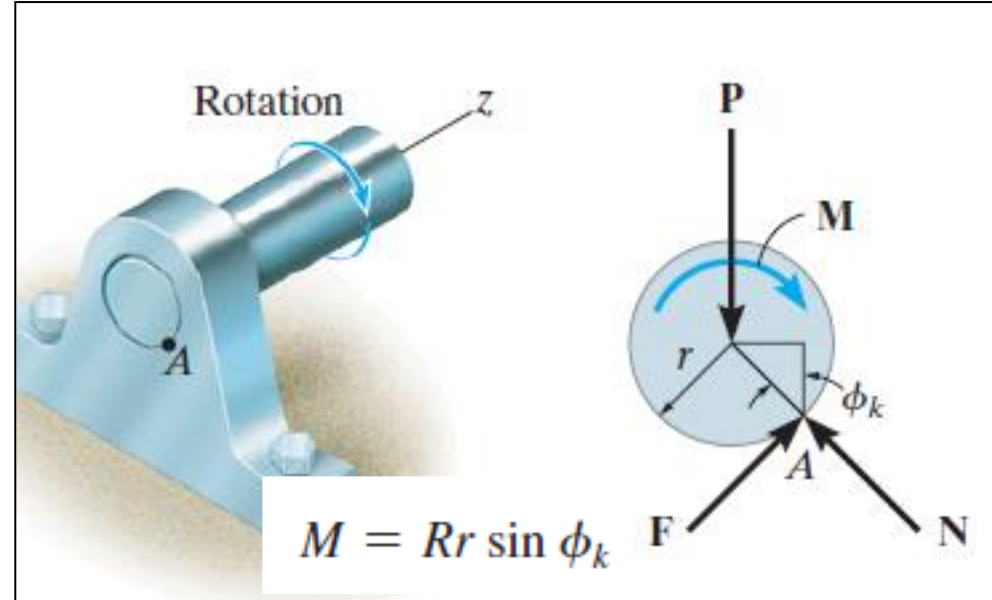
Chapter review (contd.)

Journal Bearings

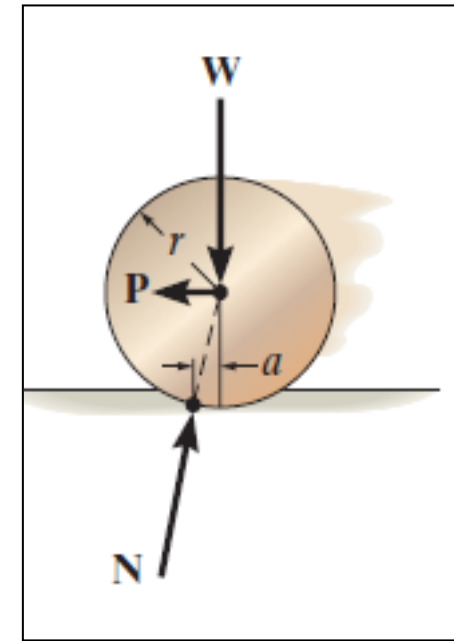
When a moment is applied to a shaft in a nonlubricated or partially lubricated journal bearing, the shaft will tend to roll up the side of the bearing until slipping occurs. This defines the radius of a friction circle, and from it the moment needed to turn the shaft can be determined.

Rolling Resistance

The resistance of a wheel to rolling over a surface is caused by localized *deformation* of the two materials in contact. This causes the resultant normal force acting on the rolling body to be inclined so that it provides a component that acts in the opposite direction of the applied force **P** causing the motion. This effect is characterized using the *coefficient of rolling resistance*, a , which is determined from experiment.



$$P \approx \frac{Wa}{r}$$



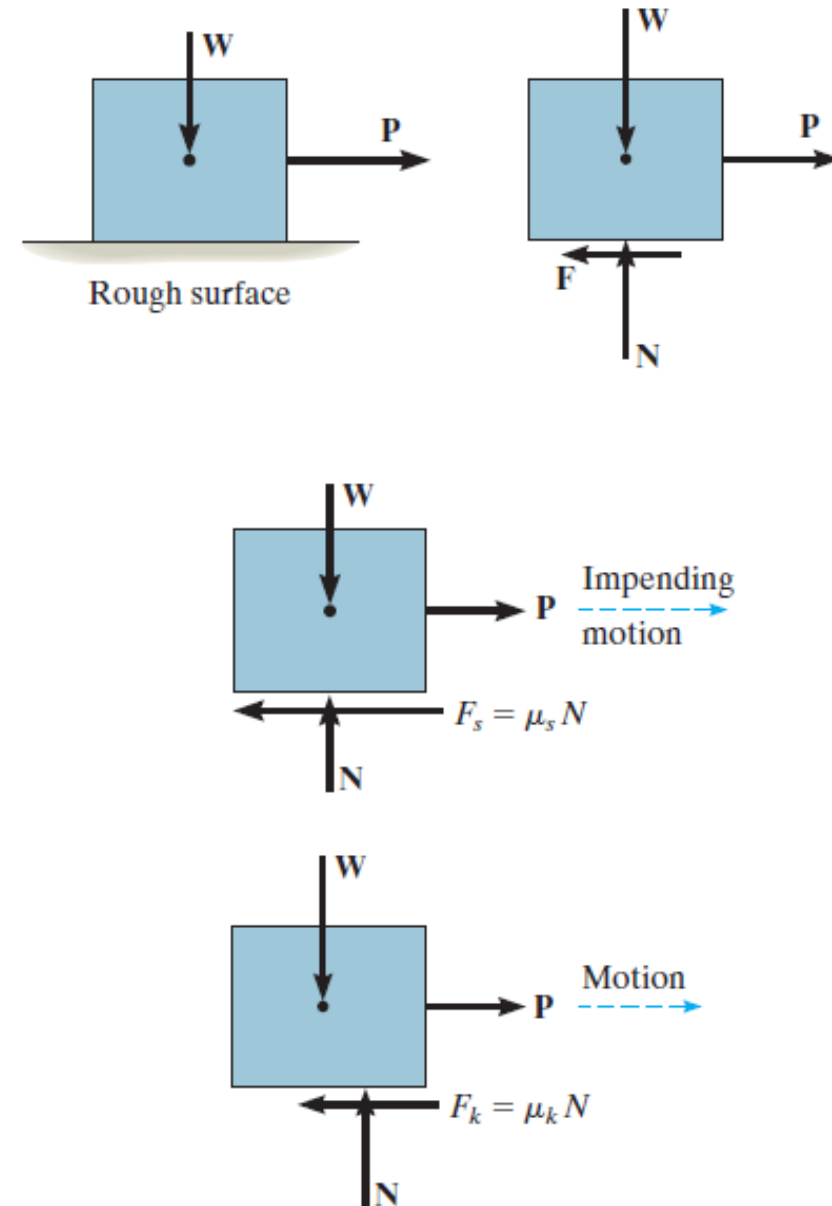
Chapter review

Dry Friction

Frictional forces exist between two rough surfaces of contact. These forces act on a body so as to oppose its motion or tendency of motion.

A static frictional force approaches a maximum value of $F_s = \mu_s N$, where μ_s is the *coefficient of static friction*. In this case, motion between the contacting surfaces is *impending*.

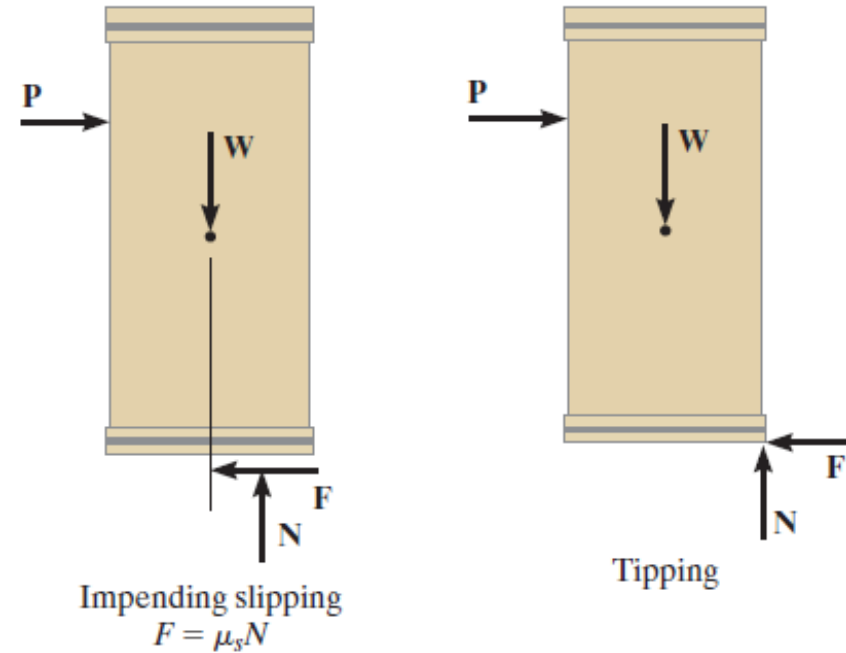
If slipping occurs, then the friction force remains essentially constant and equal to $F_k = \mu_k N$. Here μ_k is the *coefficient of kinetic friction*.



Chapter review (contd.)

The solution of a problem involving friction requires first drawing the free-body diagram of the body. If the unknowns cannot be determined strictly from the equations of equilibrium, and the possibility of slipping occurs, then the friction equation should be applied at the appropriate points of contact in order to complete the solution.

It may also be possible for slender objects, like crates, to tip over, and this situation should also be investigated.



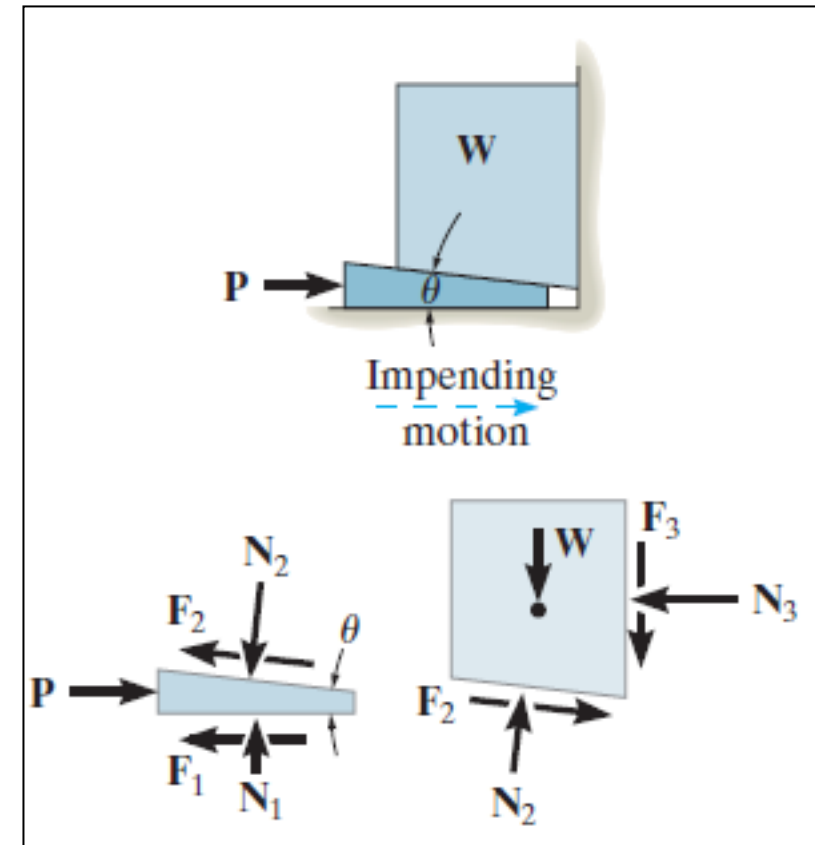
Chapter review (contd.)

Wedges

Wedges are inclined planes used to increase the application of a force. The two force equilibrium equations are used to relate the forces acting on the wedge.

An applied force **P** must push on the wedge to move it to the right.

If the coefficients of friction between the surfaces are large enough, then **P** can be removed, and the wedge will be self-locking and remain in place.



$$\begin{aligned}\Sigma F_x &= 0 \\ \Sigma F_y &= 0\end{aligned}$$

Chapter review (contd.)

Screws

Square-threaded screws are used to move heavy loads. They represent an inclined plane, wrapped around a cylinder.

The moment needed to turn a screw depends upon the coefficient of friction and the screw's lead angle θ .

If the coefficient of friction between the surfaces is large enough, then the screw will support the load without tending to turn, i.e., it will be self-locking.

$$M = rW \tan(\theta + \phi_s)$$

Upward Impending Screw Motion

$$M' = rW \tan(\theta - \phi_s)$$

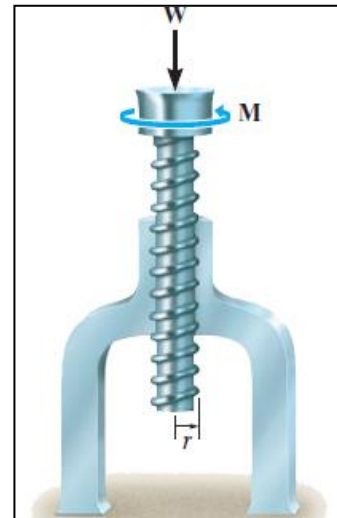
Downward Impending Screw Motion

$$\theta > \phi_s$$

$$M'' = rW \tan(\phi_s - \theta)$$

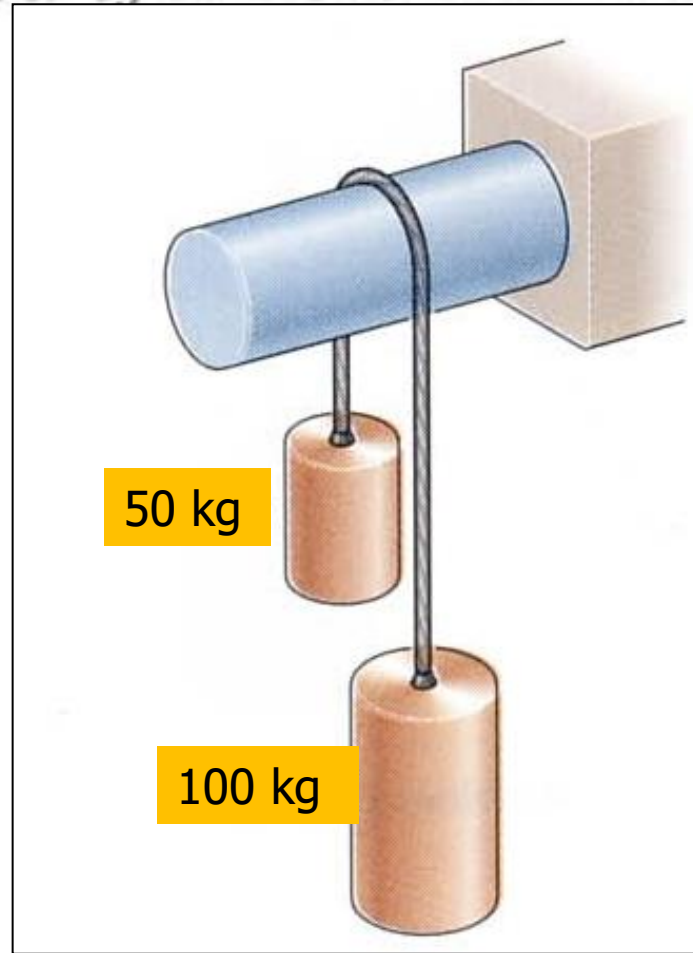
Downward Screw Motion

$$\phi_s > \theta$$

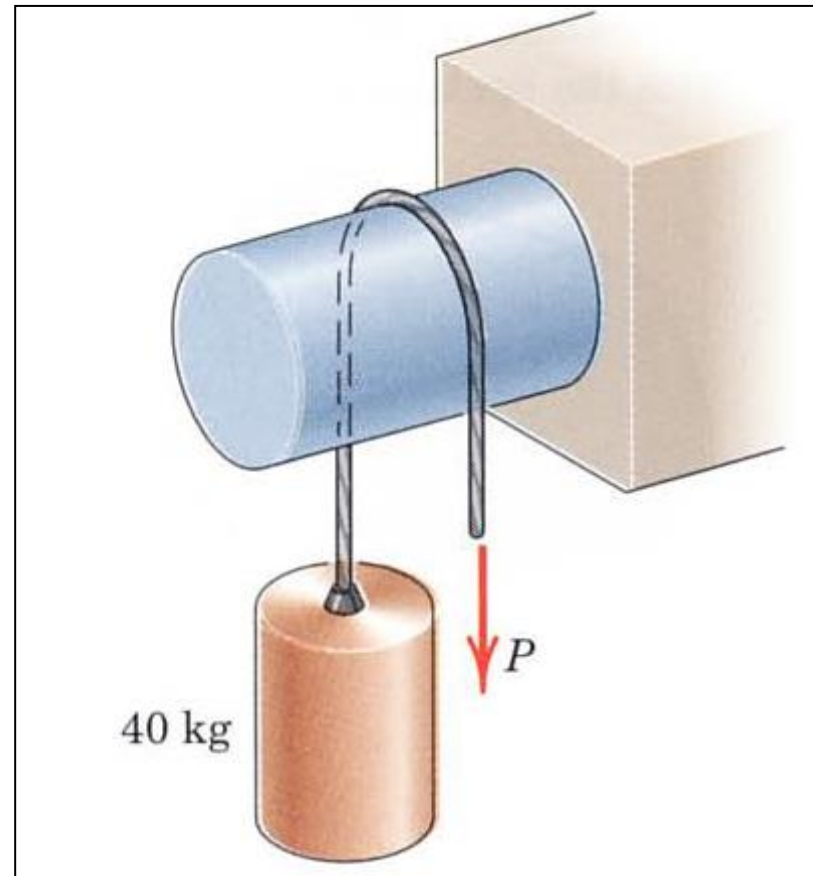


Assignments

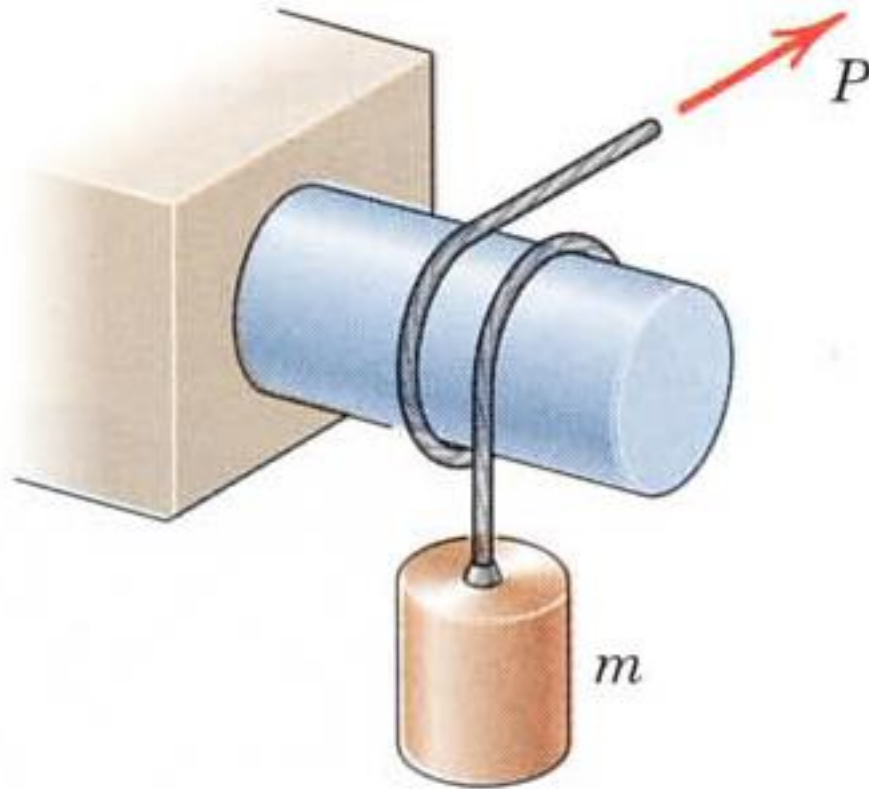
What is the minimum coefficient of friction μ between the rope and the fixed shaft which will prevent the unbalanced cylinders from moving?



- Determine the force P required to (a) raise and (b) lower the 40-kg cylinder at a slow steady speed. The coefficient of friction between the cord and its supporting surface is 0.30.



6/90 A force $P = mg/6$ is required to lower the cylinder at a constant slow speed with the cord making $1\frac{1}{4}$ turns around the fixed shaft. Calculate the coefficient of friction μ between the cord and the shaft.



Problem 6/90

End